

Harry Nguyen

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SUMMARY

ML systems engineer with production experience in GPU-aware inference serving (vLLM/PagedAttention, KV cache memory optimization), distributed backend (gRPC, Protobuf), and low-level performance engineering (C++ lock-free structures, 96% latency reduction at Chrome scale). Primary engineer for backend, cloud infra, and ML serving on a production team. Seeking to apply hardware-aware ML systems thinking at infrastructure scale.

EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

EXPERIENCE

TiMoto AI | Software Engineer

Sep 2025 – Present

- Led backend, cloud infrastructure, and ML serving for a 3-person engineering team – designed, deployed, and operated **distributed production systems** end-to-end
- Selected **vLLM/PagedAttention** for concurrent LLM inference serving: applied GPU memory paging model (non-contiguous KV cache blocks) to eliminate fragmentation under concurrent load – validated with load testing at Nx production concurrency; deployed with continuous batching achieving **zero OOM failures** at production traffic, **sub-50ms p99**
- Designed and maintained **gRPC inter-service layer**; traced circular lock-acquisition conflict causing concurrent-call deadlock and redesigned call sequencing – **100% evaluation success rate** at **sub-50ms p99**
- Architected multi-AZ ECS Fargate with **Terraform IaC**, CloudWatch observability, and circuit breaker pattern – **99.9% uptime**, **44% cost reduction** (\$40–60/month); auto-rollback on health check failure
- Maintained distributed production systems across gRPC, PostgreSQL, and ML serving layers – triaged incidents via root-cause analysis, documented runbooks, on-call rotation

Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers – analyzed cache line and memory access patterns for lock-free transport; selected Protobuf over custom serialization for schema evolution; shipped to Chrome stable serving **3B+ active users** at **sub-50ms p99**, **10K+ req/sec**
- Profiled settings navigation as p99 bottleneck at 1,200ms; designed **lock-free concurrent trie** eliminating mutex contention – verified linearizability proof before shipping; **96% latency reduction**, zero production regressions
- Architected event-driven TypeScript/React system with observer pattern decoupling UI state propagation – accelerated feature delivery **68%** across **25K+ lines** of Chromium at **95% test coverage**
- Delivered design documents and code reviews adopted by senior Chrome engineers; shipped to production branch

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on AWS – JWT over session auth for horizontal scalability; auto-scaling supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned with PostgreSQL indexing achieving **sub-100ms** for 10,000+ records

PROJECTS

Pulumi – Contributor | *Go, TypeScript, IaC, Multi-cloud*

github.com/pulumi

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning – under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer – studied linearizability guarantees and correctness under concurrent operations and partial failures

SKILLS

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, GPU memory management, LoRA/QLoRA, PyTorch, LLM-as-a-judge evaluation, Inference SLO benchmarking

Distributed Systems: gRPC, Protocol Buffers, Circuit breakers, Exactly-once semantics, Fault tolerance, Kafka, Multi-AZ, Event-driven architecture

Languages: C++, Python, Go, TypeScript, Java, JavaScript, SQL, Bash, Rust

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, K8s, Docker, CI/CD, CloudWatch, Prometheus, Grafana, GCP

Frameworks & Databases: Django, FastAPI, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor